



Brief article

The capacity of visual short-term memory within and between hemifields

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Abstract

Visual short-term memory (VSTM) and attention are both thought to have a capacity limit of four items [e.g. Luck, S. J., & Vogel, E. K. (1997). The capacity of visual working memory for features and conjunctions. *Nature*, 309, 279–281; Pylyshyn, Z. W., & Storm, R. W. (1988). Tracking multiple independent targets: evidence for a parallel tracking mechanism. *Spatial Vision*, 3, 179–197.]. Using the multiple object visual tracking paradigm (MOT), it has recently been shown that twice as many items can be simultaneously attended when they are separated between two visual fields compared to when they are all presented within the same hemifield [Alvarez, G. A., & Cavanagh, P. (2004). Independent attention resources for the left and right visual hemifields (Abstract). *Journal of Vision*, 4(8), 29a.]. Does VSTM capacity also increase when the items to be remembered are distributed between the two visual fields? The current paper investigated this central issue in two different tasks, namely a color and spatial location change detection task, in which the items were displayed either in the two visual fields or in the same hemifield. The data revealed that only memory capacity for spatial locations and not colors increased when the items were separated between the two visual fields. These findings support the view of VSTM as a chain of capacity limited operations where the spatial selection of stimuli, which dominates in both spatial location VSTM and MOT, occupies the first place and shows independence between the two fields.

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Most mental activities entail the short-term storage and manipulation of information in memory. Within the visual modality, the number of items that can be simultaneously encoded and stored appears to be limited to around four (Irwin, 1992; Kahneman, Treisman, & Gibbs, 1992; Luck & Vogel, 1997; Sperling, 1960). Classically, the storage capacity of visual short-term memory (VSTM) is assessed using the change detection paradigm (Phillips, 1974) where two arrays of multiple items are presented briefly one after the other, separated by a short interval. Participants have to detect a change between the two arrays. In general performance in such a task is nearly perfect when the arrays contain up to four items, but decreases significantly if more than four items are presented.

Interestingly, a similar capacity limit of four objects has also been encountered in studies investigating the rapid visual selection of information. Evidence comes from the study of multiple object visual tracking (MOT) (Pylyshyn & Storm, 1988; Scholl & Pylyshyn, 1999; Yantis, 1992) where participants have to attentionally track a number of moving identical items in a display of identical distractors. In a typical MOT task, participants are shown a display of identical items. A subset of these items is then flashed several times to assign them as targets. All the items then start to move independently and randomly within the display and participants are instructed to keep track of the targets. Finally, after a period of time the items stop moving and participants have to point out which items were the targets. Pylyshyn and Storm (1988) found that participants were able to successfully track up to four items.

The observation of a four-item limitation in both change detection and MOT studies has led some authors to propose that the capacity limit of VSTM and attention may simply be one and the same, or at least different aspects of the same process (Cowan, 1995; Rensink, 2002). Only the objects in a visual scene that are attended get stored in VSTM. There is indeed more and more evidence that suggests a close relationship between VSTM and attention. For example, it has been found that attended items are better stored in VSTM than unattended items (Palmer, 1988; Schmidt, Vogel, Woodman, & Luck, 2002; Scott-Brown & Orbach, 1998; Woodman, Vecera, & Luck, 2003). This is consistent with other findings that have shown that when individuals attend to a particular object or event, they often fail to report the appearance of irrelevant, unexpected items (e.g. Becklen & Cervone, 1983; Neisser & Becklen, 1975). This has been interpreted as *inattention blindness*, a failure to represent unattended items (Mack & Rock, 1998).

In addition to the same capacity limit of four items, both visual attention and VSTM have been shown to be contralaterally organized. Psychophysiological (Heinze, Luck, Mangun, & Hillyard, 1990; Luck & Hillyard, 1994) and neuroimaging (LaBerge, 1990) investigations, have demonstrated that visual attention effects are often most evident in the hemisphere contralateral to the hemifield to which attention is directed. Likewise, there is some evidence indicating the existence of a contralateral organization for visual memories (see the review by Gratton, 1998). For example, using a recognition task in humans, Gratton and colleagues (Gratton, Corballis, & Jain, 1997) have found a recognition advantage for line patterns presented in the same hemifield at study and test compared to patterns presented in different hemifields, suggesting a basic difference in memory content between the two hemispheres. Some authors have gone further to suggest that totally independent resources may exist for attentional selection in the two hemifields. For example, Luck and colleagues (Luck, Hillyard, Mangun, & Gazzaniga, 1989) found that

split-brain patients could process bilateral displays faster than unilateral displays, indicating not only a contralateral organization, but also independent attentional capacities in the left and right hemispheres. Likewise, using a MOT paradigm in healthy participants, [Alvarez and Cavanagh \(2004\)](#) have recently demonstrated that twice as many items can be simultaneously tracked when they are split between the two hemifields as when they are all presented within the same hemifield.

According to the view of a strict link between capacity limits of attention and VSTM, we should also expect independent capacity limits in VSTM for the left and right hemifield, such that twice as many items can be simultaneously encoded and maintained when they are separated across the two hemifields compared to when they are all displayed in the same hemifield. To my knowledge, the question of capacity limits in VSTM within and between hemifields has never been directly investigated in the literature. The current experiments were designed to address this issue.

The change detection paradigm was used to estimate the storage capacity of VSTM within and between hemifields. This paradigm has been shown to provide relatively uncontaminated estimates of VSTM capacity, since change detection performance has been shown to be unaffected by a verbal memory load ([Luck & Vogel, 1997](#)). Of main interest was if VSTM capacity increases when the items are split between the two hemifields compared to when they are all displayed in the same hemifield.

1. Experiment 1: color VSTM

The goal of this first experiment was to examine whether a greater number of colors can be simultaneously encoded into VSTM when they are separated across two visual fields compared to when they are all displayed within one field.

1.1. Method

1.1.1. Participants

Data were included from 13 naïve participants (10 females; mean age = 24; range 19–30). All had normal (self-reported) or corrected-to-normal visual acuity and color vision.

1.1.2. Stimuli and procedure

In all experiments, visual stimuli were displayed on the 17-in. screen of a 733 MHz PC running E-Prime 1.1 software. Responses were collected from the computer keyboard.

On each trial, participants were shown a central fixation point (a small white cross subtending $0.48^\circ \times 0.48^\circ$ at viewing distance of 60 cm) and four white frames (subtending $4.95^\circ \times 4.95^\circ$ each) separated vertically and horizontally by 7.12° (center to center; see [Fig. 1](#)). The whole display subtended $11.95^\circ \times 11.95^\circ$, and the screen background was gray (127 on red, green, and blue phosphors). There were nine (3×3) virtual sectors within each of the four frames, such that each sector was separated by 1.62° (center to center).

After viewing the four unfilled frames for 1000 ms, participants were presented with several small colored squares (subtending $0.76^\circ \times 0.76^\circ$ each) randomly displayed within one or two frames, for 200 ms. Each square occupied a particular sector in the frame

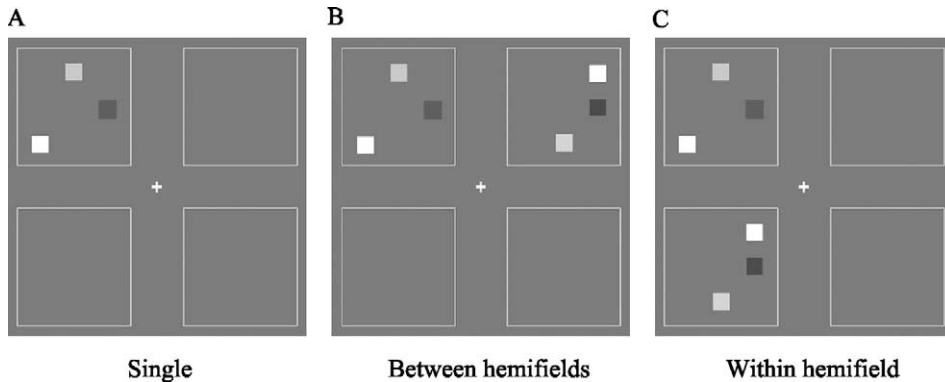


Fig. 1. Samples of displays used in the color VSTM experiment. In the *single condition* (A), the stimuli appeared in one frame only; in the *between hemifields condition* (B), the stimuli appeared in two frames from different hemifields; in the *within hemifield condition* (C), the stimuli appeared in two frames from the same hemifield. The different gray levels represent different colors.

and could be one of six possible colors (green, red, blue, yellow, pink, and turquoise). There was no repetition of color within the same display.

After a 1000 ms blank interval, the squares reappeared (test display) and remained present until a response was given. Participants indicated by a key-press, as accurately and as quickly as possible, whether the colors of the squares in the test display were the same as, or different from the ones from the sample display. In 50% of the trials, the colors of the squares from the sample and test displays were the same, whereas in the other 50% of the trials, one square changed its color from the sample to the test display. When the color of a square changed in the test display, the changed color was one of the remaining colors not previously allocated to any squares from the same frame. The next trial started 1000 ms later.

The squares could appear in two frames from different hemifields (above or below of fixation) (*between hemifields condition*) or in two frames from the same hemifield (left or right of fixation) (*within hemifield condition*). In order to examine whether a complete independence of the two visual fields can be observed, the squares could also appear, in a third condition, in one frame only (*single condition*). If each hemifield shows a total independence for color VSTM, performance should be similar in the single and between hemifields conditions. There were also three different set sizes: two, three, or four squares could appear within a particular frame. The three conditions consisted of 120 trials each (40 per set size), presented in random order in a single session lasting approximately 30 m. At the beginning of the session, the instructions were given to the participants, who completed 18 practices. Eye movements were not monitored, but participants were encouraged to keep their eyes focused on the central fixation cross during each trial.

1.2. Results and discussion

In the data analysis, A' was calculated as a measure of the participant's sensitivity (Grier, 1971). The means of A' for each condition and for each set size are plotted in

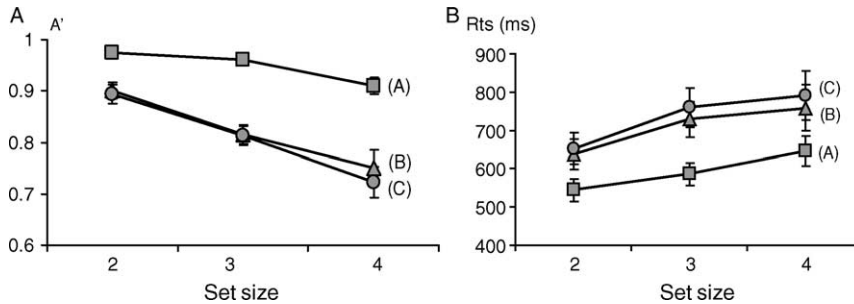


Fig. 2. Results (A' and RTs) of the color VSTM experiment. The single, between and within conditions correspond in the graph to labels (A), (B), and (C), respectively. Error bars represent standard errors of the means. Note that the set size involves the number of items *per* frame, such as a set size of 4 involves four items for the single condition but eight items for the between and within conditions.

Fig. 2A. The set size $\times$ condition ANOVAs (repeated measures) revealed a significant main effect of condition [$F(2, 24) = 49.491$; $MSE = 0.0046$, $P < .001$], a significant main effect of set size [$F(2, 24) = 58.823$; $MSE = 0.0028$, $P < .001$], and a significant interaction between condition and set size [$F(4, 48) = 3.651$; $MSE = 0.0031$, $P < .05$]. Post hoc analyses (Tukey HSD tests) showed that accuracy was significantly higher in the single condition than in the two other conditions ($P < .05$ in set size 2; $P < .001$ in set sizes 3 and 4), which did not differ from each other. The set size effect was significant in the within and between hemifields conditions, with better performance at smaller set sizes ($P < .05$), but not in the single condition. A set size $\times$ condition ANOVAs (repeated measures) was also conducted on the correct response latencies (Fig. 2B) and revealed a significant main effect of condition [$F(2, 24) = 31.664$; $MSE = 6976$, $P < .001$], a significant main effect of set size [$F(2, 24) = 21.572$; $MSE = 6777$, $P < .001$], but no interaction. Post hoc analyses (Tukey HSD tests) showed faster responses in the single condition than in the two other conditions ($P < .005$), which did not differ from each other. Performance at set size 4 was also poorer than performance at set sizes 2 and 3 ($P < .005$), which tended to differ from each other ($P = .053$).

These data show that performance was equivalent for the within and between hemifields conditions, which is not consistent with the existence of independent VSTM storage resources in the two visual hemifields. Why does this color VSTM task not show hemifield independence whereas other capacity-limited processes, such as attentional processing (Alvarez & Cavanagh, 2004; Luck et al., 1989), do? One possible explanation may be that two different types of information are used in these tasks. Attentional tracking deals with spatial information whereas color short-term memory deals with identity information. It is now well accepted that two separate streams of information processing in the visual system exist for processing spatial and identity information (Ungerleider & Mishkin, 1982), and it has also been suggested that short-term memory systems dedicated to spatial and identity information can be functionally and anatomically dissociated (e.g. Courtney, Ungerleider, Keil, & Haxby, 1996; Logie & Marchetti, 1991; Mecklinger & Müller, 1996; Smith et al., 1995). It is therefore possible that only spatial information is

processed independently in the two fields. To address this possibility, short-term memory for spatial locations within and between hemifield was examined in a second experiment.

2. Experiment 2: spatial location VSTM

The purpose of this experiment was to examine whether more spatial locations can be stored in VSTM when they are presented in two visual fields compared to when they are all displayed within the same hemifield.

2.1. Method

2.1.1. Participants

The same 13 participants from Experiment 1 took part in this 30-min study. Half of them performed the Experiment 2 prior to the Experiment 1.

2.1.2. Stimuli and procedure

The stimuli and procedure were the same as Experiment 1, except that the small squares that appeared in the frames during the sample and test displays were all white (Fig. 3). Participants had to indicate whether the locations occupied by the squares remained the same between sample and test display. In 50% of the trials, one of the squares changed its location, moving to one of the 3×3 locations that was unoccupied in the first array.

2.2. Results and discussion

The means of A' for each condition and for each set size are plotted in Fig. 4A. The set size $\times$ condition ANOVA (repeated measures) revealed a significant main effect of condition [$F(2, 24) = 56.428$; $MSE = 0.0025$, $P < .001$], a significant main effect of set size [$F(2, 24) = 15.668$; $MSE = 0.0015$, $P < .001$], and a significant interaction between condition and set size [$F(4, 48) = 5.725$; $MSE = 0.0016$, $P < .001$]. The effect of condition

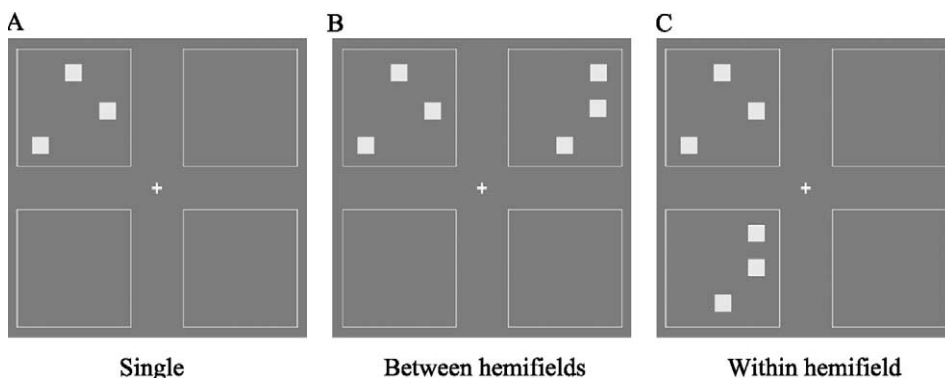


Fig. 3. Samples of displays used in the spatial location VSTM experiment.

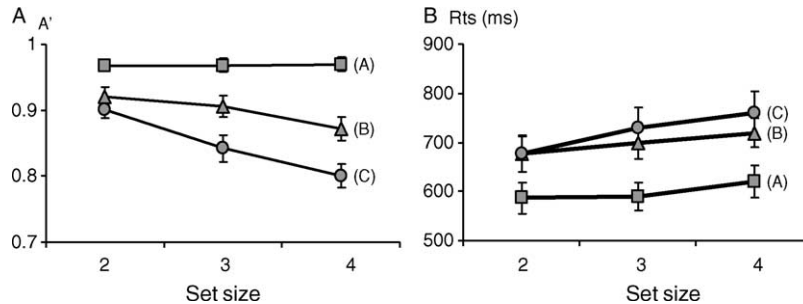


Fig. 4. Results (A' and RTs) of the spatial location VSTM experiment. The single, between and within conditions correspond in the graph to labels (A), (B), and (C), respectively. Error bars represent standard errors of the means. Again, the set size involves the number of items per frame, such as a set size of 4 involves four items for the single condition but eight items for the between and within conditions.

was significant in set sizes 3 and 4 only, as shown by post hoc analyses (Tukey HSD tests), with better performance in the single condition than in the two other conditions ($P < .01$ in set size 3; $P < .001$ in set size 4), and with better performance in the between hemifields than in the within hemifield condition ($P < .01$ in set size 3; $P < .005$ in set size 4). The set size effect was significant in the within hemifield condition only, with better performance in set sizes 2 and 3 than in set size 4 ($P < .05$). A set size $\times$ condition ANOVAs (repeated measures) was also conducted on the correct response latencies (Fig. 4B) and revealed a significant main effect of condition [$F(2, 24) = 42.69$; $MSE = 3925$, $P < .001$], a significant main effect of set size [$F(2, 24) = 8.376$; $MSE = 3215$, $P < .005$], but no interaction. Post hoc analyses (Tukey HSD tests) showed faster responses in the single condition than in the two other conditions ($P < .005$), which did not differ from each other. Finally, response time was longer with set size 4 than set size 2 in the within hemifield condition ($P < .005$).

The critical finding from this experiment is that spatial location VSTM can benefit from the distribution of the locations between the two visual fields, indicating that a certain degree of hemifield independence exists for VSTM capacity.

3. General discussion

The primary purpose of the present study was to examine whether the number of objects that can be simultaneously encoded in VSTM can be increased by presenting the objects in two visual fields rather than one. The results showed that only memory for spatial locations and not memory for colors benefits from the distribution of the objects across two fields. Why does memory for spatial locations and not color memory show this degree of hemifield independence? While the current experiments cannot directly answer to this question, it is useful to consider two speculative explanations offered by Alvarez and Cavanagh (2004) to account for their finding that attentional tracking shows hemifield independence whereas visual search (Luck et al., 1989) and memory storage (Duncan et al., 1999) do not.

First, these authors suggest that hemifield independence might be observed in tasks that involve attention to spatial location (such as a tracking task, for instance), but not in tasks that involve attention to identity information (such as most visual search and memory tasks). At first glance, this explanation in terms of the type of information involved could easily account for the present findings: a certain degree of hemifield independence was observed in the spatial location VSTM task where attention had to be directed to spatial locations, whereas no indication of hemifield independence was found in the color task where attention to identity was required. Nevertheless, this explanation appears to be insufficient in view of the present data. In contrast to the attentional tracking of moving objects, in which a complete independence of the two visual fields was observed (Alvarez & Cavanagh, 2004), spatial location VSTM did not show total hemifield independence as performance was still better in the single than in the between hemifields condition (see Fig. 4A). Thus, the type of information (spatial versus identity) may not be the only factor determining whether independence between the two visual fields is observed.

The second possibility offered by these authors is that hemifield independence can only be found in the first stage of a sequence of capacity limited operations, that of stimulus selection, whereas later processing stages, such as identification and memory storage, may deal with information from both fields. Attentional tracking shows hemifield independence since target selection is primarily involved, most visual search and memory tasks do not show independence between the two fields since subsequent stages, of identification and storage, are primarily involved. According to this view, spatial location VSTM involves both stimulus selection and memory storage, but is especially dominated by limits imposed at the selection stage which give rise to a certain degree of hemifield independence. Here, the partial independence of the two fields may be explained by the fact that the memory storage stage could slightly erase the hemifield independence resulting from the selection stage. In contrast, color VSTM may be dominated by processing stages such as identification and identity storage that could fully erase any hemifield independence resulting from earlier stages.

To sum up, the experiments reported here provide the first compelling demonstration that VSTM capacity can be increased by presenting objects in two rather than one visual hemifield. This effect was found for spatial locations but not colors, suggesting that the capacity of VSTM must not be seen as unitary. Rather, in accordance with Alvarez and Cavanagh (2004), VSTM capacity may be the result of a chain of capacity limited operations, where only the process of stimulus selection, that precedes identification and storage, shows independence between the two visual fields.

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